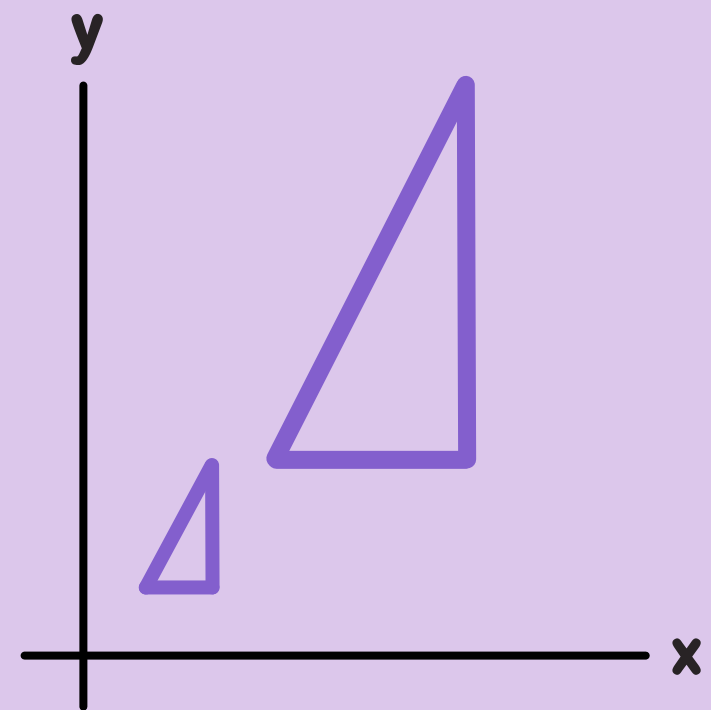
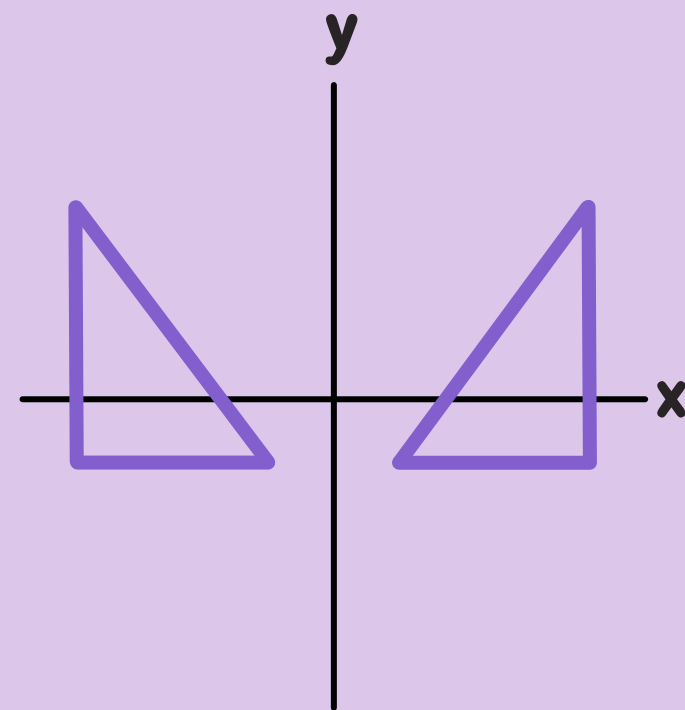
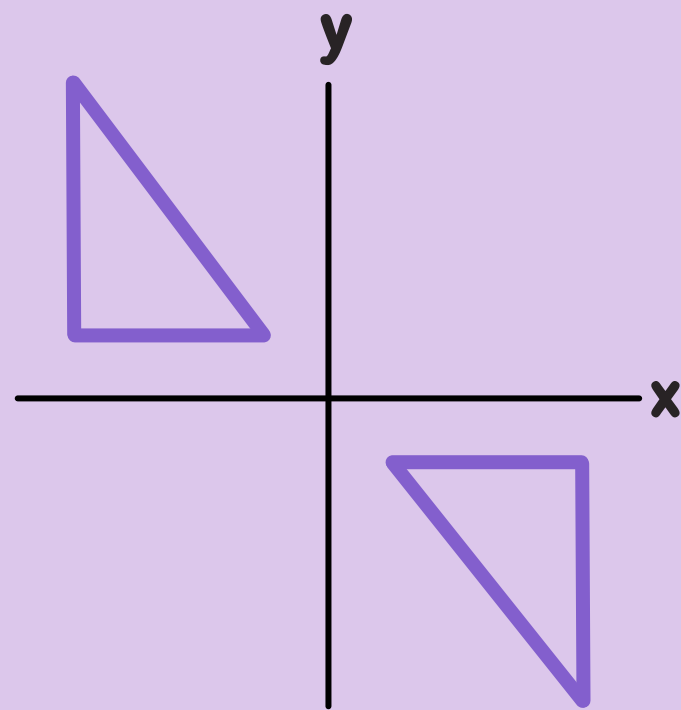
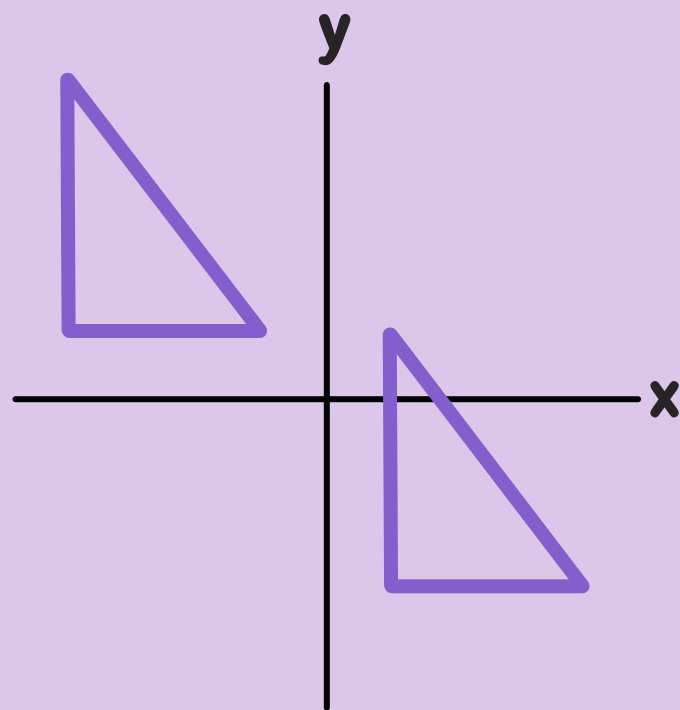


TRANSFORMATIONS

GRADE 7



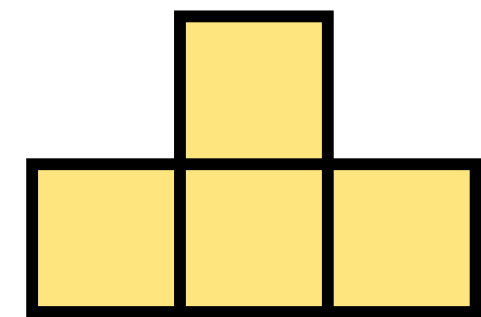
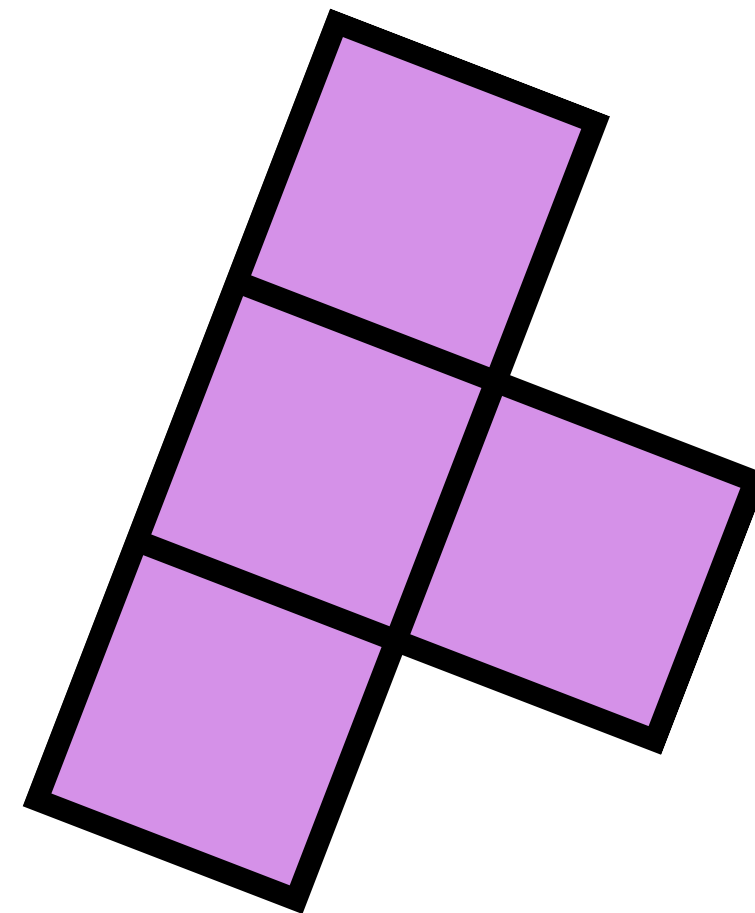
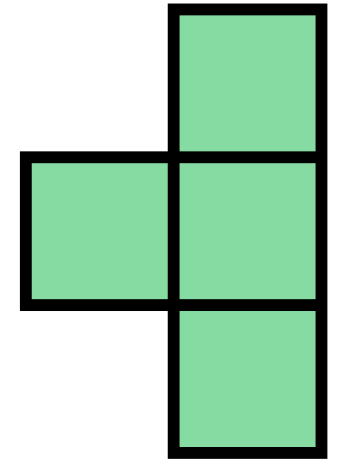
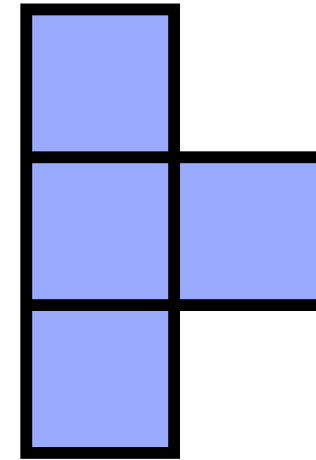
LEARNING GOAL



**STUDENTS ARE ABLE TO
USE COORDINATE RULES
TO TRANSFORM IMAGES**

INTRODUCTION

Geometric transformations are operations that change the position, size, or orientation of shapes while preserving their basic properties.



KEYWORDS



CONGRUENCE

Two figures are congruent if they are the identical in shape and size

SIMILARITY

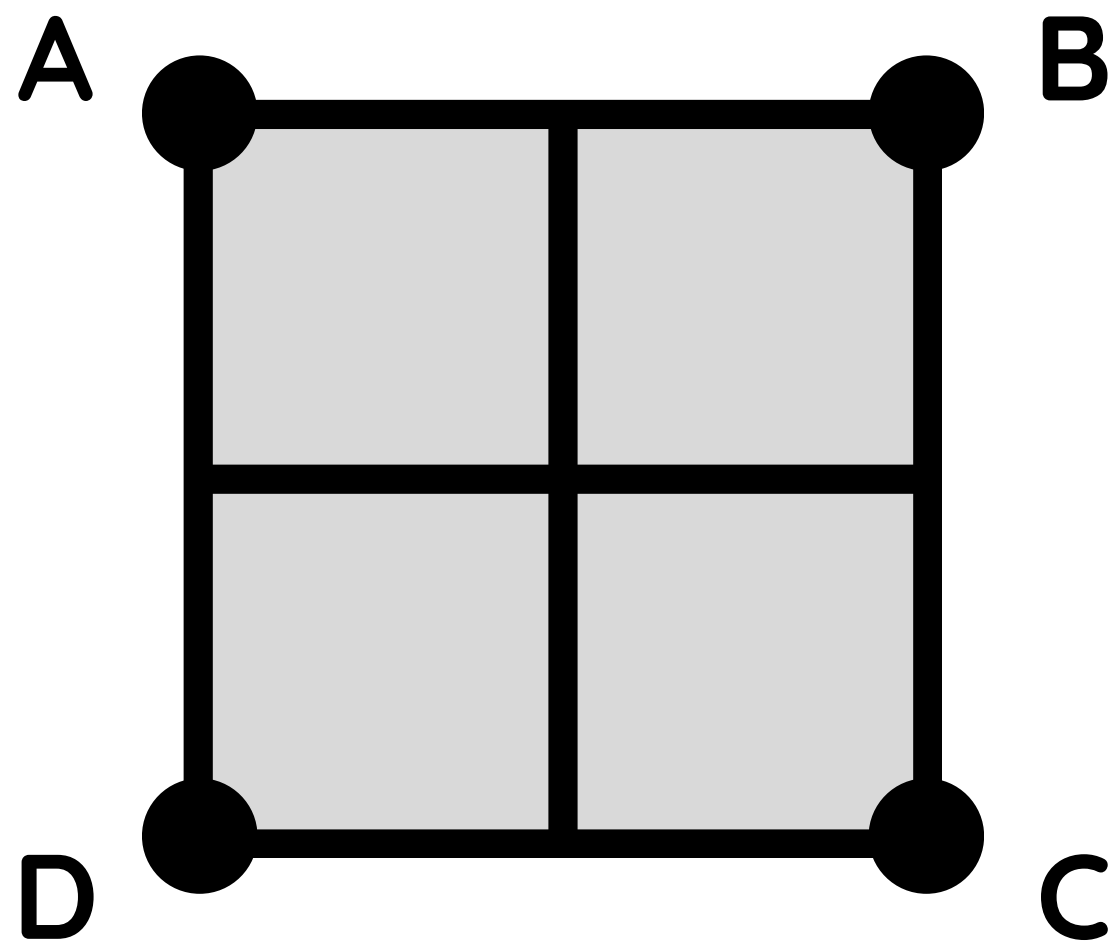
Two figures are similar if they are both the same shape but different sizes, where corresponding sides are in proportion.



NOTATION

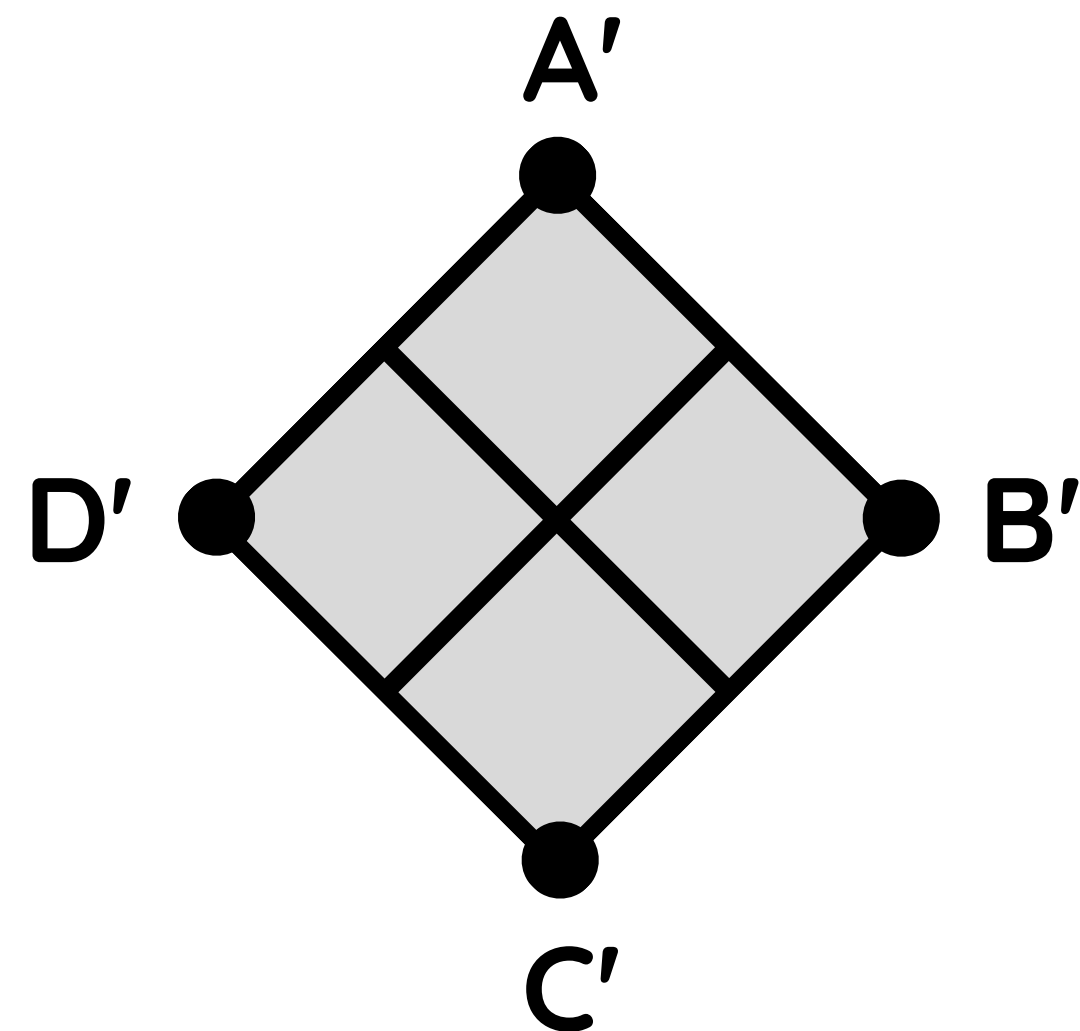
Pre-image

before transformation(s)



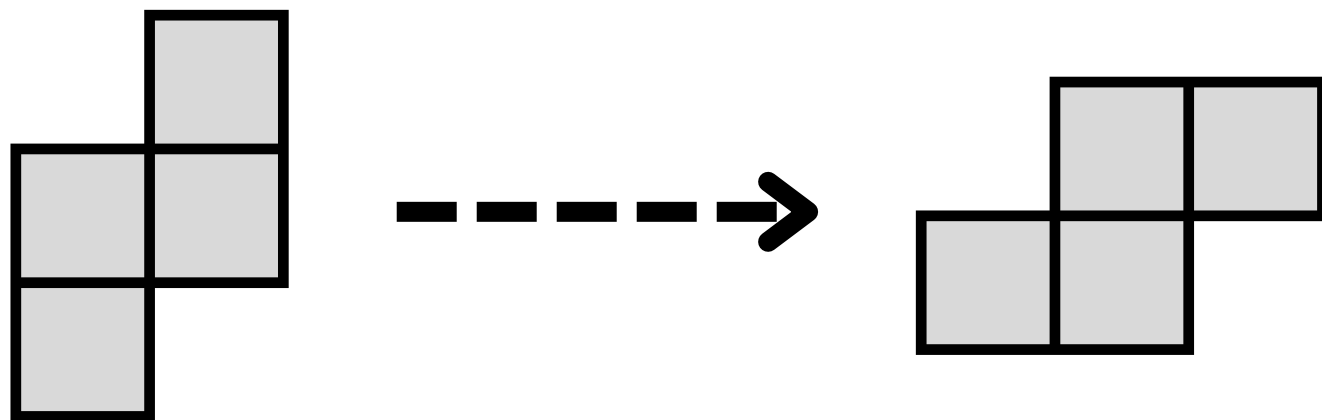
Image

after transformation(s)



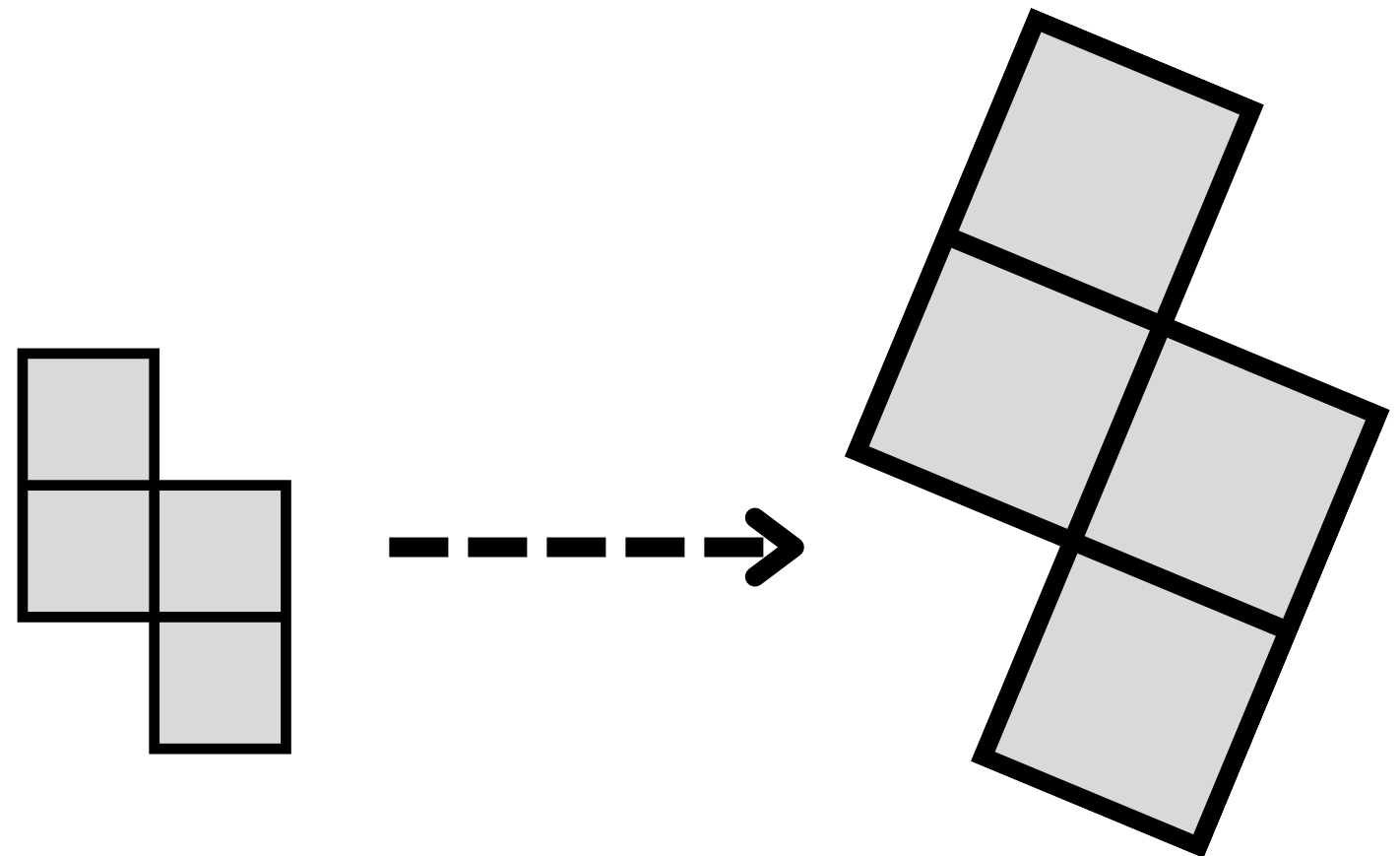
RIGID

- Preserves size and shape
- Also called isometries

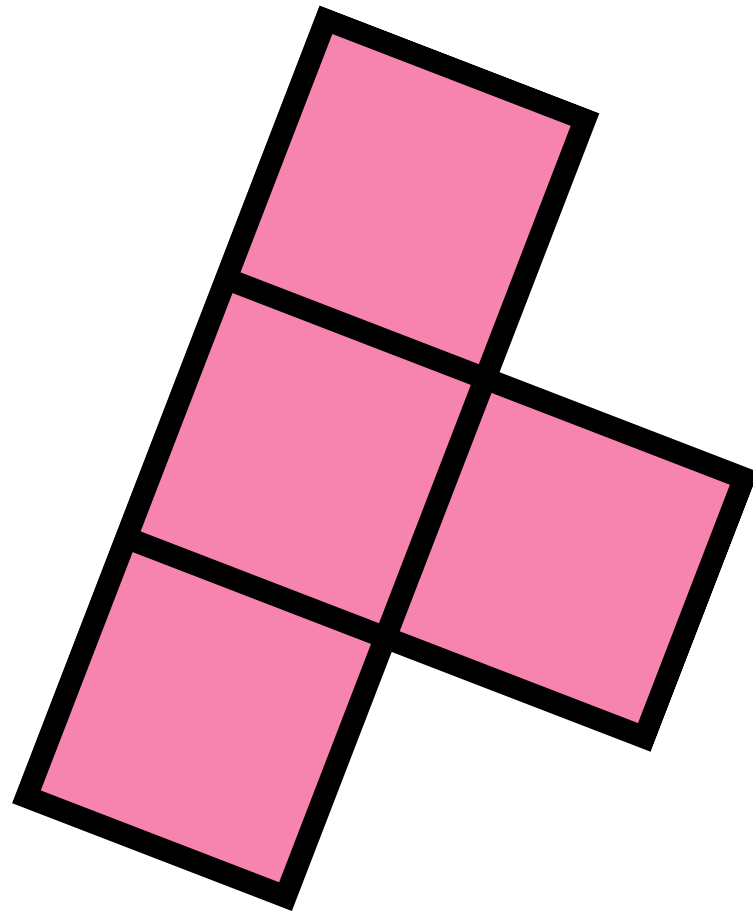
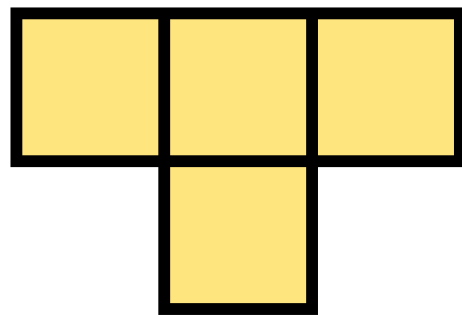
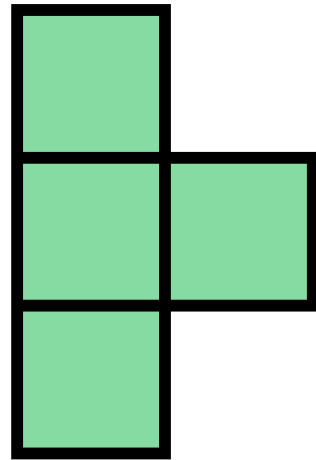
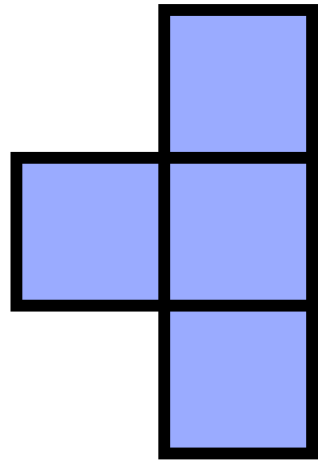


NON-RIGID

- Preserves shape but not size



TYPES OF TRANSFORMATIONS



Rigid

- Translation - "slide"
- Reflection - "flip"
- Rotation - "turn"

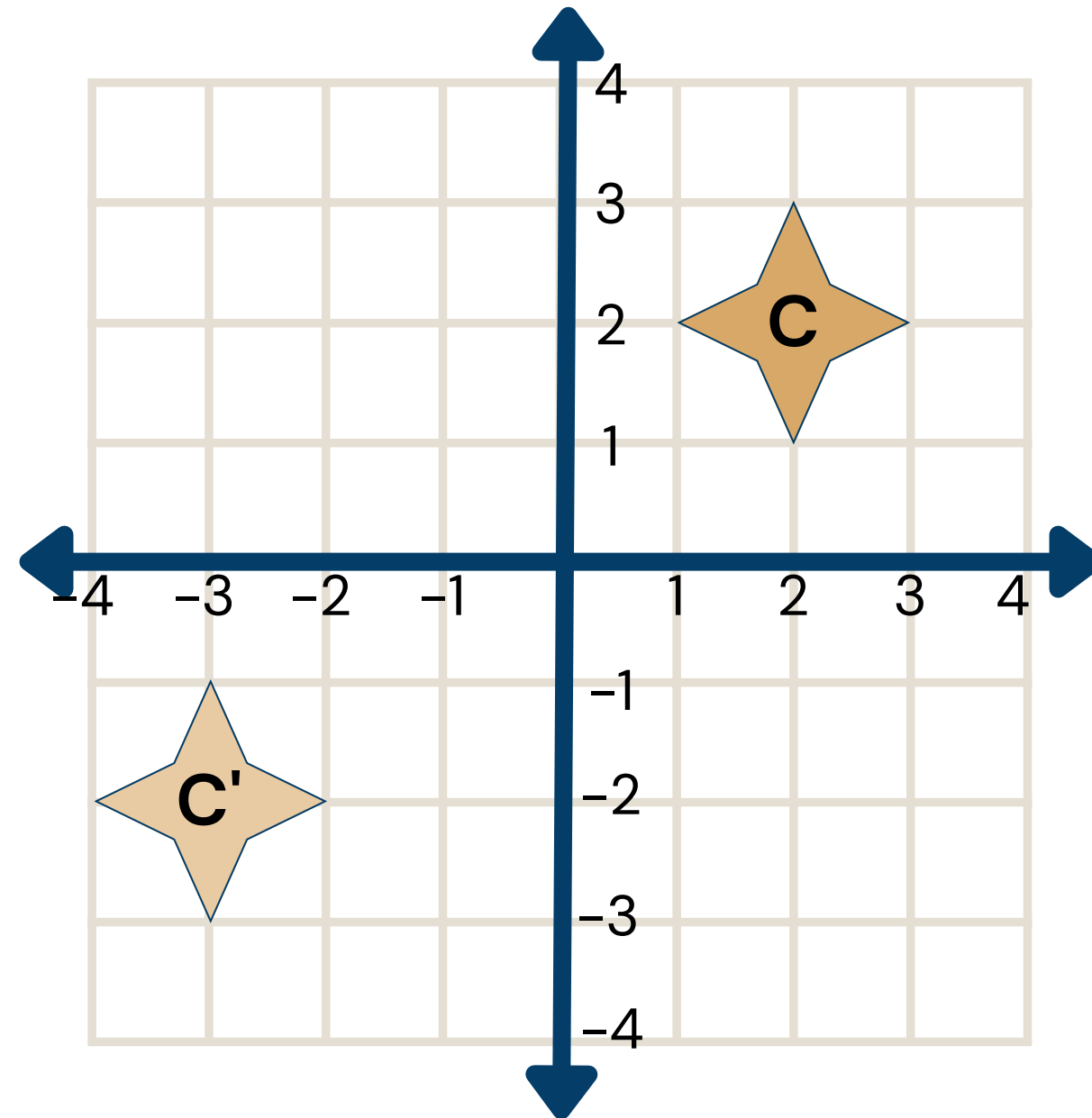
Non - Rigid

- Dilation - "stretch or shrink"

TRANSLATION

A translation moves a shape left or right and/or up or down.

The shape and size remain the same, therefore both original and translated shapes are congruent.

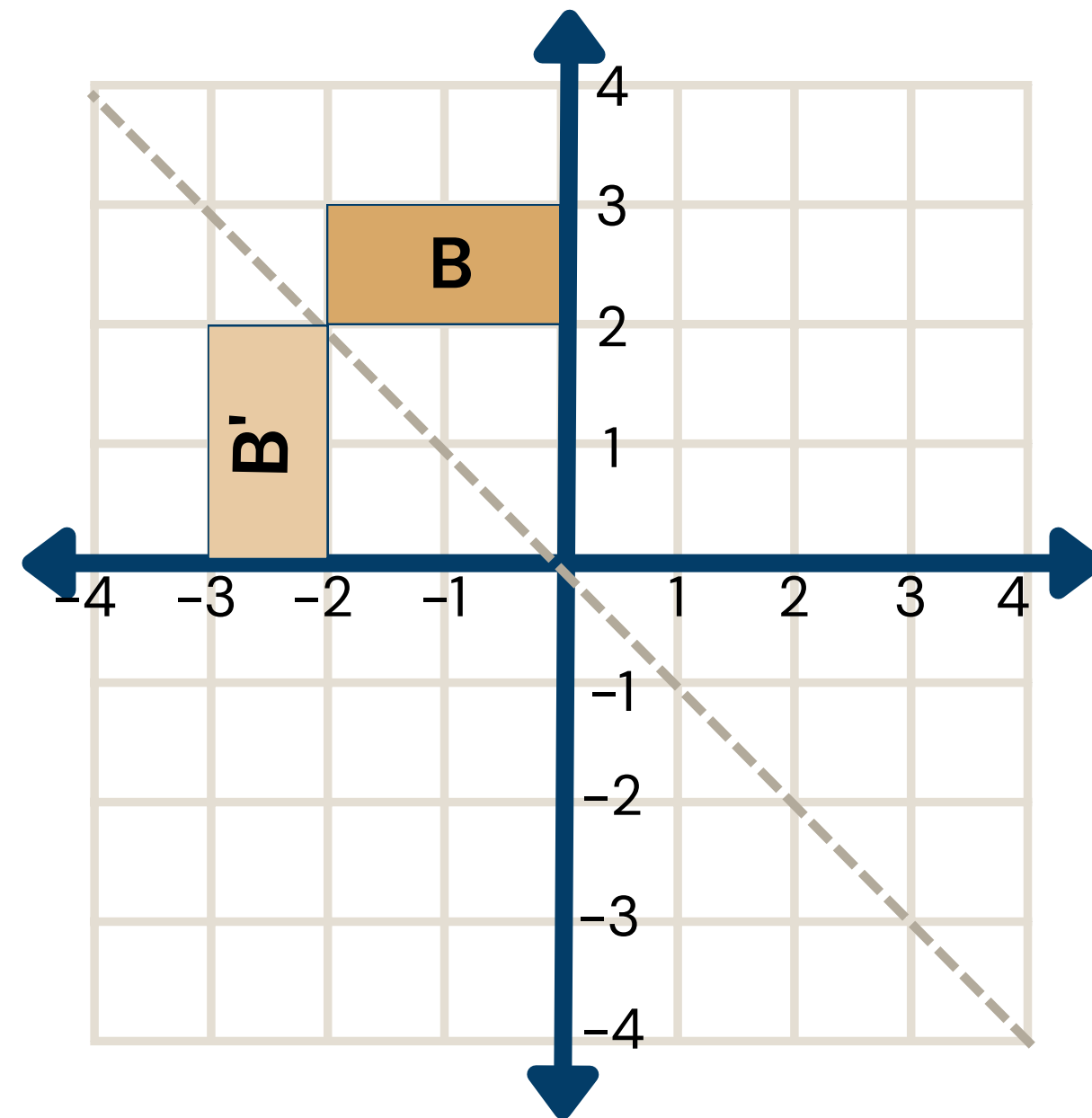


Shape C has been translated 5 units left
and 4 units down

REFLECTION

A reflection flips an object to produce a mirror image.

The shape and size remain the same, therefore both original and reflected shapes are congruent.

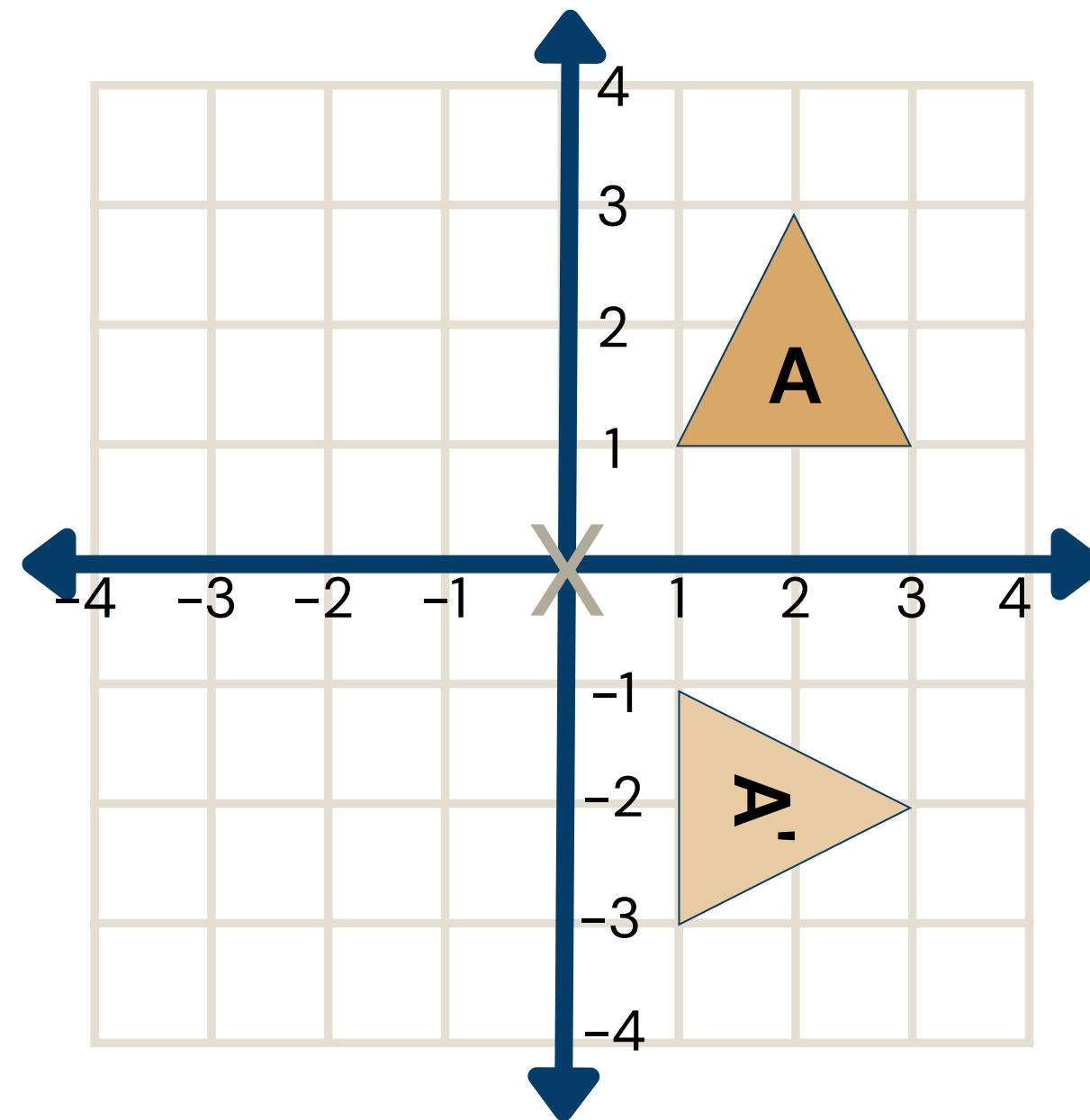


Shape B has been reflected in the line $y = -x$ to produce image B'

ROTATION

A rotation turns an object through a given angle and direction around a center point.

The shape and size remain the same, therefore both original and rotated shapes are congruent.

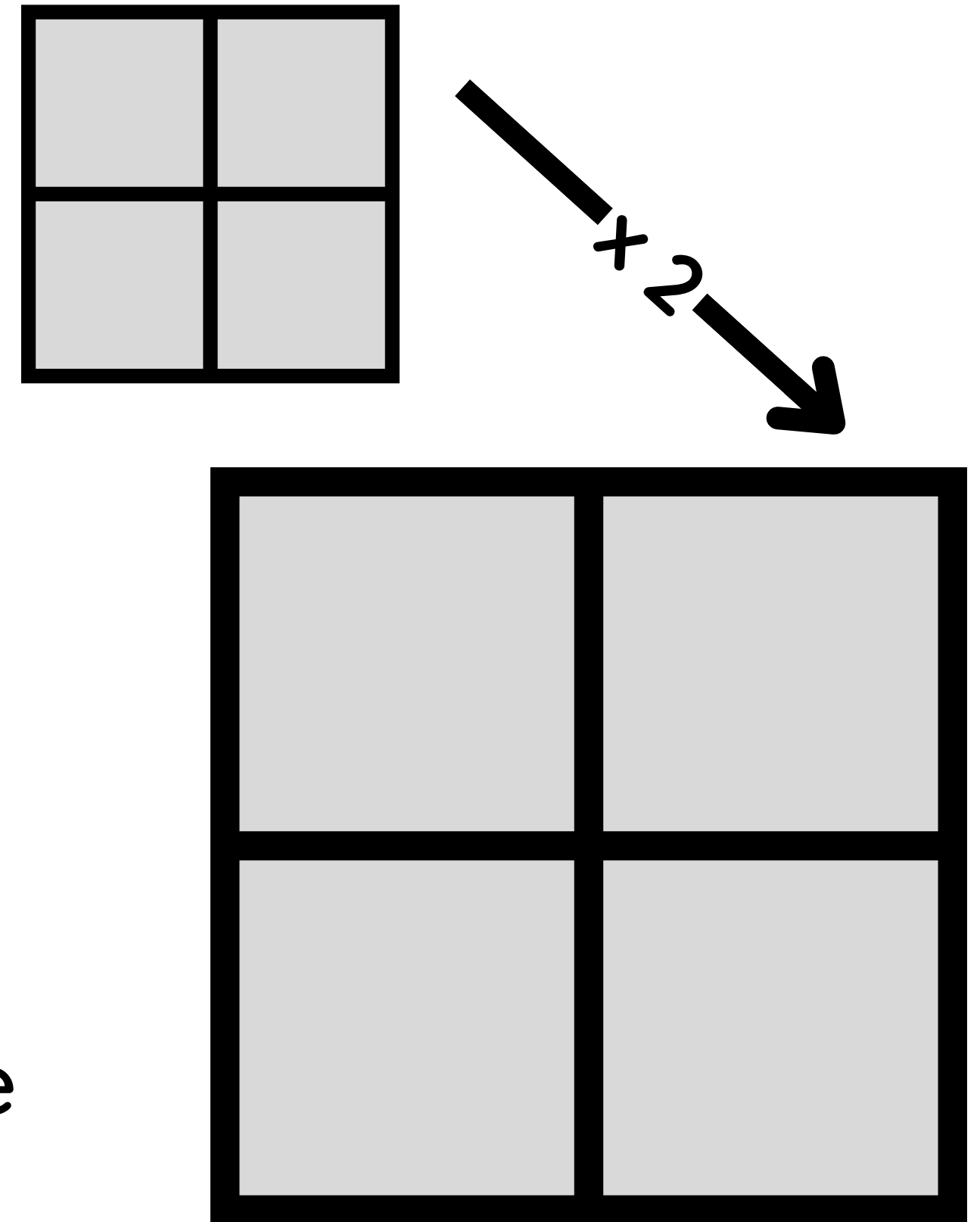


Shape A has been rotated 90° clockwise about the origin

DILATION

A rigid transformation in which a shape made bigger or smaller by a given factor called the scale factor.

A dilation that creates a larger image is called an enlargement. A dilation that creates a smaller image is called a reduction.



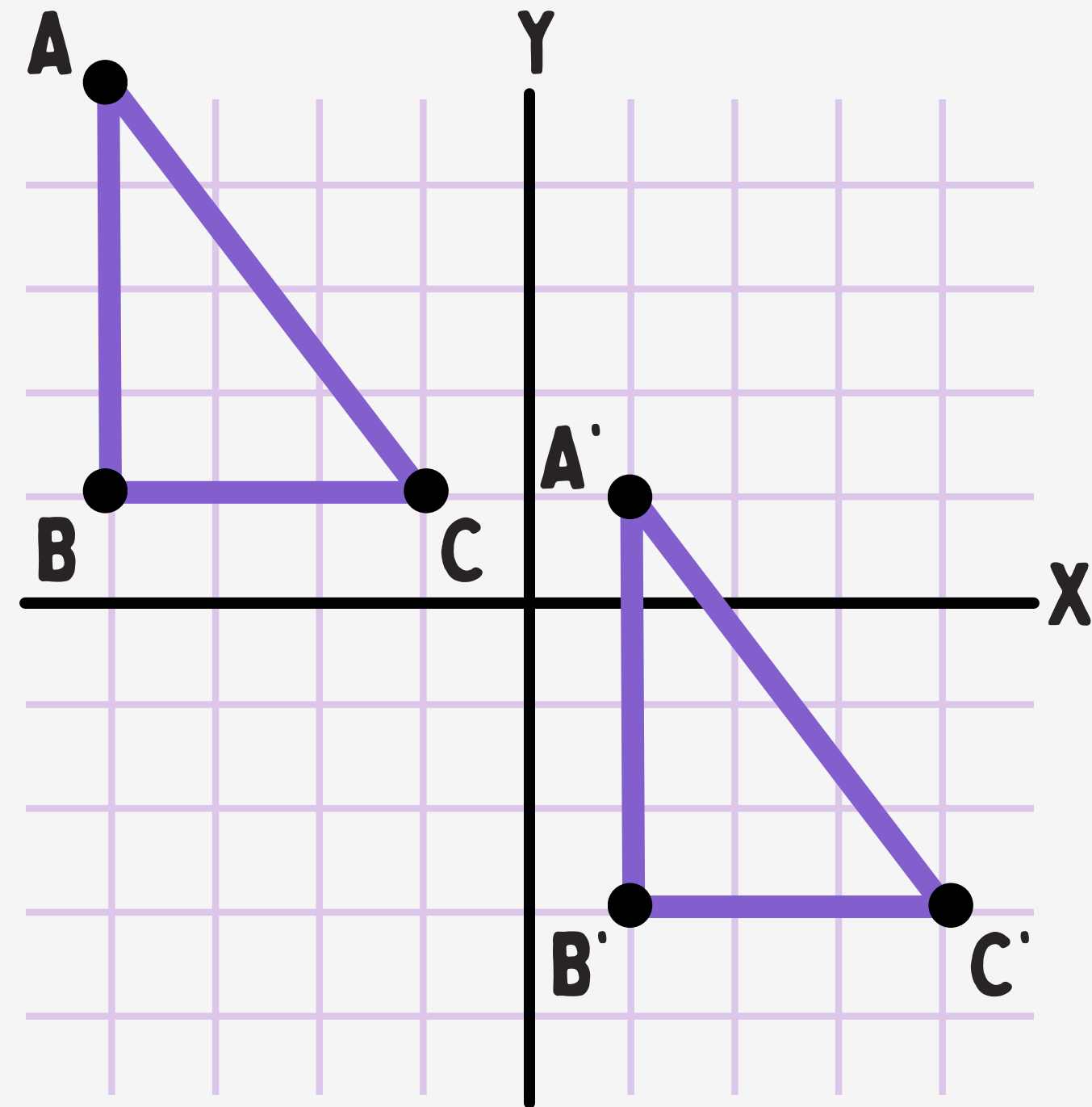
TRANSLATIONS

WHAT DO YOU NOTICE HAPPENS TO
THE X AND Y COORDINATES WHEN
ABC IS TRANSLATED 5 UNITS RIGHT
AND 4 UNITS DOWN TO GET A'B'C'?

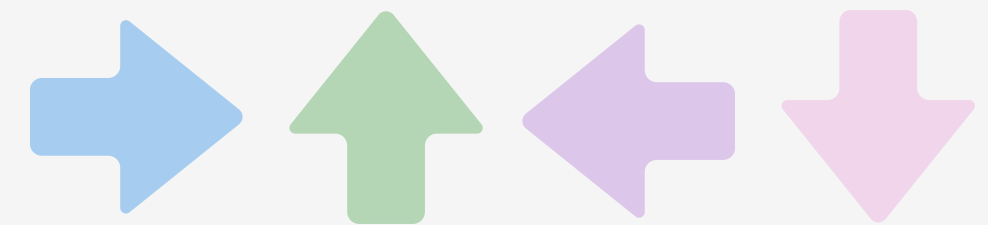
$$A(-4,5) \rightarrow A'(1,1)$$

$$B(-4,1) \rightarrow B'(1,-3)$$

$$C(-1,1) \rightarrow C'(4,-3)$$



TRANSLATION RULES



$(X,Y) \rightarrow (X+H,Y+K)$	H UNITS RIGHT, K UNITS UP
$(X,Y) \rightarrow (X+H,Y-K)$	H UNITS RIGHT, K UNITS DOWN
$(X,Y) \rightarrow (X-H,Y+K)$	H UNITS LEFT, K UNITS UP
$(X,Y) \rightarrow (X-H,Y-K)$	H UNITS LEFT, K UNITS DOWN

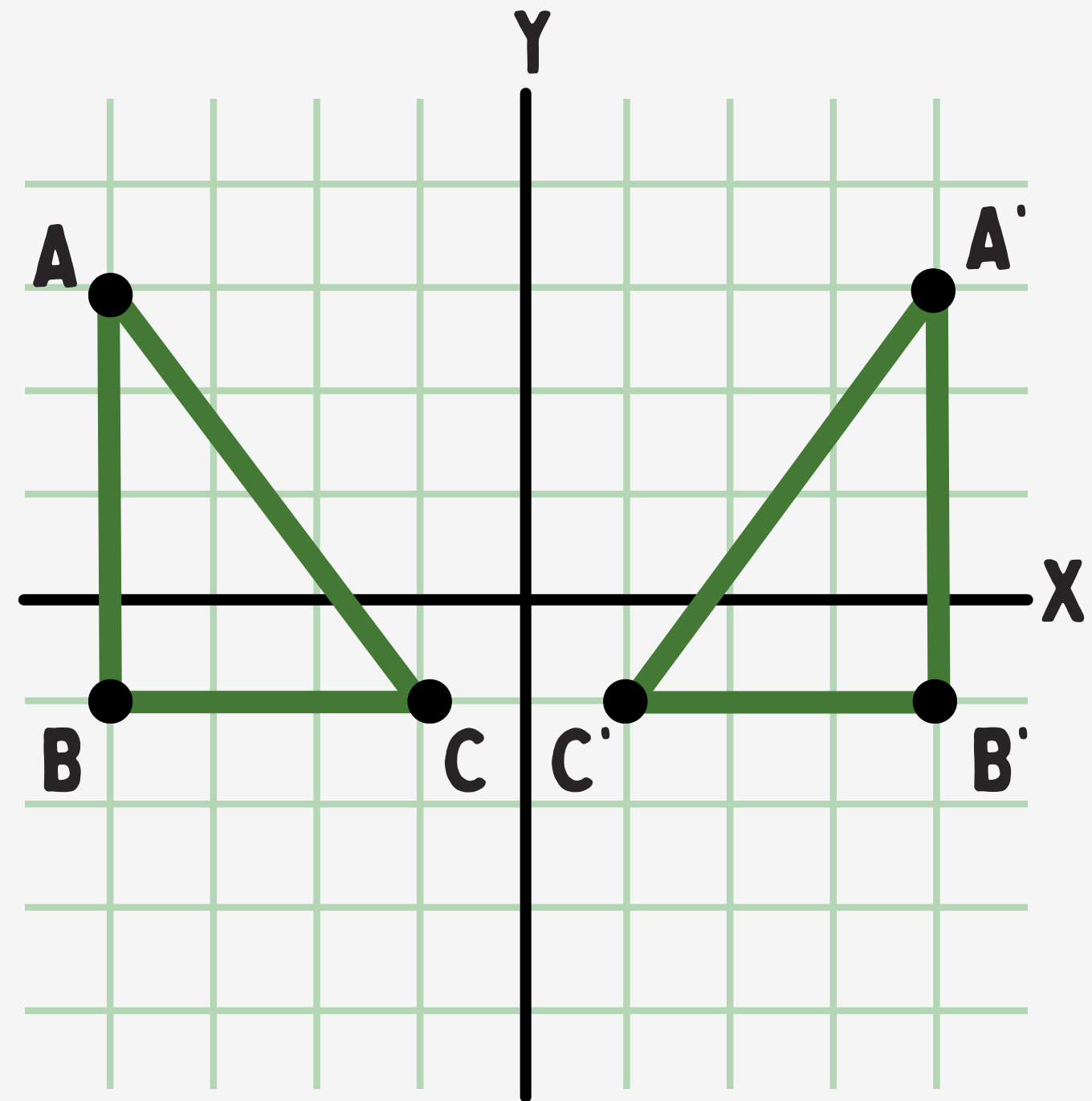
REFLECTIONS

WHAT DO YOU NOTICE HAPPENS TO THE X AND Y COORDINATES WHEN ABC IS REFLECTED OVER THE Y-AXIS TO GET A'B'C'?

$$A(-4,3) \rightarrow A'(4,3)$$

$$B(-4,-1) \rightarrow B'(4,-1)$$

$$C(-1,-1) \rightarrow C'(1,-1)$$



REFLECTION RULES



$(X,Y) \rightarrow (X,-Y)$	REFLECTION OVER THE X -AXIS
$(X,Y) \rightarrow (-X,Y)$	REFLECTION OVER THE Y -AXIS
$(X,Y) \rightarrow (Y,X)$	REFLECTION OVER THE LINE $Y=X$
$(X,Y) \rightarrow (-Y,-X)$	REFLECTION OVER THE LINE $Y=-X$

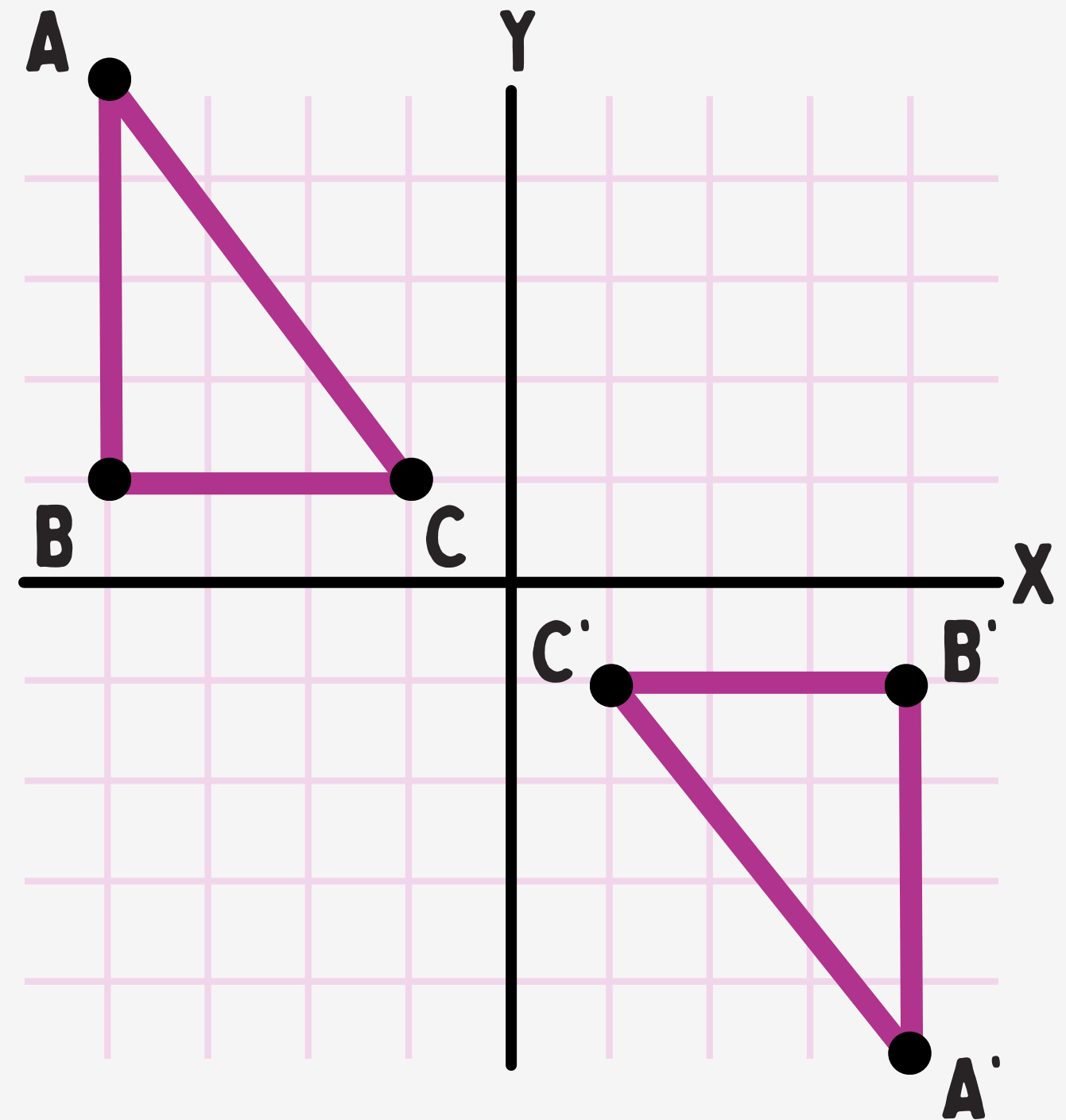
ROTATIONS

WHAT DO YOU NOTICE HAPPENS TO
THE X AND Y COORDINATES WHEN
ABC IS ROTATED 180 DEGREES TO
GET A'B'C'?

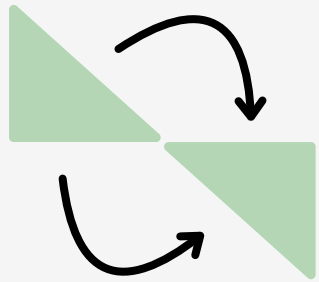
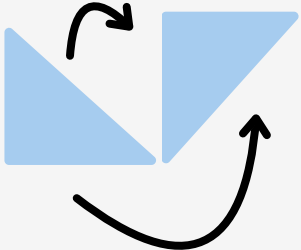
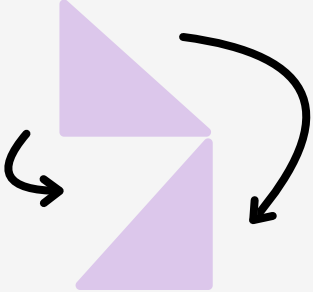
$$A(-4,5) \rightarrow A'(4,-5)$$

$$B(-4,1) \rightarrow B'(4,-1)$$

$$C(-1,1) \rightarrow C'(1,-1)$$



ROTATION RULES ABOUT THE ORIGIN

$(X,Y) \rightarrow (-X,-Y)$	180° CLOCKWISE OR COUNTERCLOCKWISE	
$(X,Y) \rightarrow (Y,-X)$	90° CLOCKWISE 270° COUNTERCLOCKWISE	
$(X,Y) \rightarrow (-Y,X)$	270° CLOCKWISE 90° COUNTERCLOCKWISE	

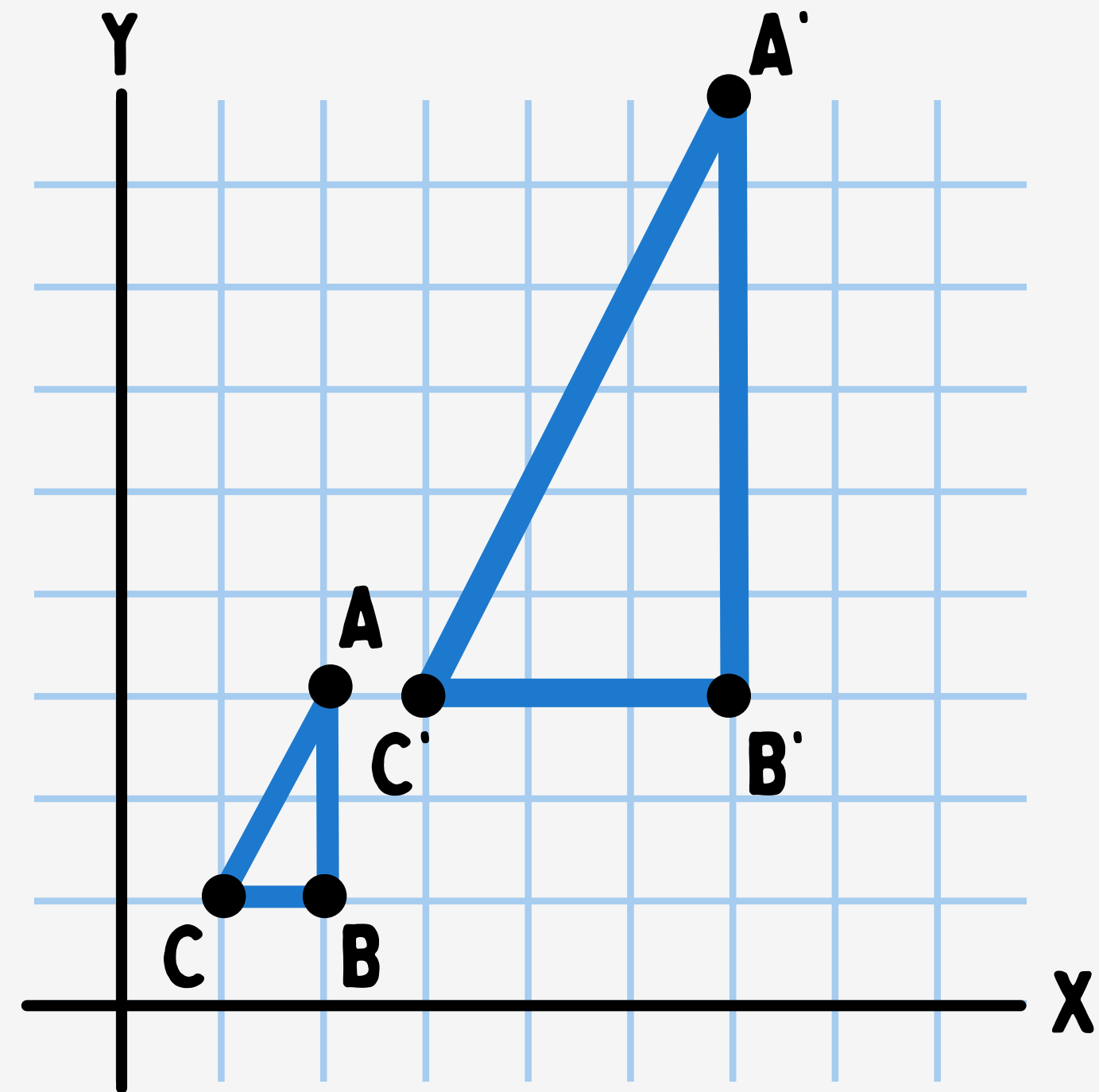
DILATIONS

WHAT DO YOU NOTICE HAPPENS TO
THE X AND Y COORDINATES WHEN
ABC IS DILATED BY A SCALE FACTOR
OF 3 TO GET A'B'C'?

$$A(2,3) \rightarrow A'(6,9)$$

$$B(2,1) \rightarrow B'(6,3)$$

$$C(1,1) \rightarrow C'(3,3)$$



DILATION RULE ABOUT THE ORIGIN

$$(X,Y) \rightarrow (KX,KY)$$

DILATED BY A SCALE FACTOR OF **K**

IF K IS BIGGER THAN 1

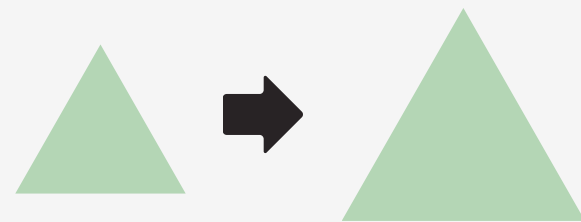


IMAGE GETS BIGGER

IF K IS BETWEEN 0 AND 1

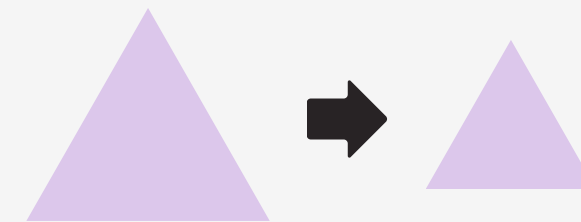
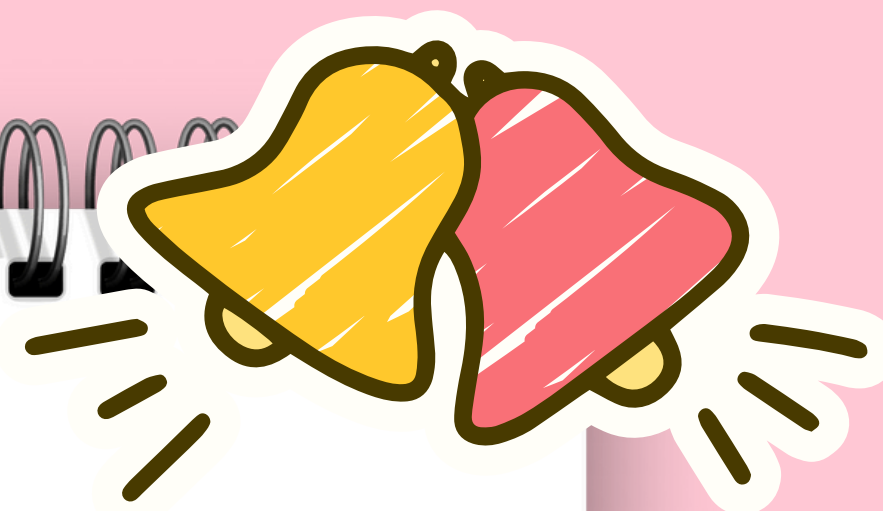


IMAGE GETS SMALLER



THANK YOU

See you next time!

